

Julian Maggin

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PROFESSIONAL SUMMARY

Senior Product Development Engineer with 13 years taking high-volume consumer hardware from concept to mass production - across premium appliances at Fisher & Paykel, plumbing and connected products at GWA Group (Caroma, Gethai), and safety-critical electromechanical products at B&D Doors / DuluxGroup. Deep expertise in 3D CAD, DFM/DFA across polymers and metals, materials selection, prototype build-and-test verification, and regulatory compliance (IP ingress sealing, AU/NZ water-contact and performance standards). Experienced managing overseas manufacturing partners across Asia, including on-site build trials.

TECHNICAL SKILLS

Design & CAD: SolidWorks (8+ yrs), Creo Parametric; detailed mechanical design, component detailing & drafting, DFM/DFA, rapid prototyping

Materials & Manufacturing: Materials/polymer selection, plastic injection moulding, sheetmetal, tooling collaboration, manufacturing engineering, production trials

Verification & Compliance: Prototype build & experimental testing, control & test plans; drop / transit / climatic testing; IP ingress sealing; AU/NZ water-contact & performance standards

Delivery & Commercial: Project management, concept-to-mass-production ownership, technical documentation & BOMs, supplier management & negotiation, cross-functional NPD

CAREER HIGHLIGHTS

- **Led the Fisher & Paykel re-brand of a Haier-platform refrigerator** - the first F&P fridge built at Haier's mass-manufacturing scale in China; redesigned the doors and handles, specified a new brushed finish, and ran on-site build trials in China so the product reads seamlessly alongside F&P's Thai-made range.
- **Designed the Caroma Contura Variable Spray Shower** handset - passed AU/NZ water-contact and performance standards first submission through detailed mechanical modelling and supplier engineering; recognised in the Good Design Awards as part of the Contura Collection.
- **Led end-to-end mechanical redesign of the B&D PE Beam** line - consolidating two legacy products into one SKU, achieving IP66 ingress sealing, cutting footprint 25%, and producing manufacturing-ready documentation for local and overseas production.
- **Designed the patent-pending dosing mechanism in the Gethai Fuse shower-infusion adaptor** (for US smart-shower brand Hai) - a check-valve interlock keeps the port sealed until the soap bulb is locked in, so the user can load or swap it with the shower running and no water escapes; seating the bulb opens the valve and diverts the stream through it to dissolve and carry the additive downstream to the handset.
- **Ran a dedicated cost-out engineering rotation** at Fisher & Paykel, delivering cost-down design and warranty-reduction across multiple refrigeration product lines.

PROFESSIONAL EXPERIENCE

Technical Engineering Recruitment Consultant

ProFound People, Sydney | Nov 2025 - Present

- Specialist engineering recruitment - scoping mechanical and product engineering briefs with technical leaders, and assessing candidate portfolios and product architectures daily.
- Actively returning to hands-on mechanical product development and manufacturing engineering.

Senior Product Development Engineer

B&D Doors / Automatic Technologies (DuluxGroup), Sydney | Oct 2024 - Nov 2025

- **Product Lead** for a new SKU consolidating two legacy products into a single design, retiring redundant tooling and streamlining the portfolio.
- **Drove DFM and cost-reduction** - 25% smaller footprint, reduced screw count, IP66 ingress sealing, and strict regulatory compliance on a safety-critical product.
- **Produced and reviewed manufacturing drawings**, tolerancing, and supplier documentation across sheetmetal brackets, moulded housings, and shared exterior parts - for both local and overseas production.
- **Improved assembly and installation** through targeted design: a single-bracket mounting system with a concave-seated sliding piece locked by one wingnut performing three clamping functions.
- **Led battery selection** (lithium vs alkaline) against shipping regulations, performance, and packaging constraints; directed transit testing validating product and packaging integrity through distribution.
- **Redesigned the PE Safety Beam** opener sub-assembly - an instrumented optical-sensor subsystem - hardening it for reliable field performance and unlocking a new market segment while retaining the established customer base.

Product Development Manager (Short-Term Contract)

Sydney Trains, Sydney | Apr 2024 - Aug 2024

- Worked within a regulated rail-infrastructure environment on in-service component reliability and supplier engagement strategy.
- Analysed component lifecycles, failure modes, and replacement frequencies; translated findings into compliant design recommendations for engineering stakeholders.
- Established the technical framework and specifications for compliant alternative components and procurement / engineering approval.

Senior Product Development Engineer

GWA Group Limited (Caroma), Sydney / Auckland | Dec 2020 - Apr 2024

- **Managed end-to-end mechanical development** of high-volume plumbing and consumer hardware from concept to release, including the Caroma Contura Variable Spray Shower and patent-pending Gethai Fuse.
- **Engineered the Gethai Fuse dosing mechanism**: paired check valves sit normally closed until the soap bulb is locked home, when ramps drive a sprung strainer that opens the top valve - tying flow to the bulb being seated, so soap loads with the shower running and nothing ejects from the side; a two-position bayonet adds a drain/rest step, and an internal pressure rib drives water through the bulb to dissolve and carry the additive downstream.
- **Led material selection and in-lab compliance testing**, ensuring products met AU/NZ water-contact and performance standards on first-round submission.
- **Directed validation testing** - drop, transit, and climatic heat-exposure - replicating worst-case Australian distribution conditions.
- **Championed DFM initiatives** that reduced unit costs and optimised supply-chain logistics across Asian manufacturing partners.
- **Directed technical and commercial negotiations** with overseas suppliers across Asia, bridging executive stakeholders, customer teams, and factory floors to secure viable, high-quality manufacturing.

Product Development Engineer

Fisher & Paykel Appliances, Auckland | Jul 2017 - Dec 2020

- **Sub-System Lead for premium installation parts and process** on the Integrated Column Refrigerator, F&P's flagship high-end product - designed installation trims, guides and brackets, managing fits and tolerances across cabinetry variation; specified, validated and tested purchased hinge hardware.
- **Led the F&P re-brand of a Haier-platform refrigerator** - the first F&P fridge built at Haier's scale in China: redesigned the doors and handles, selected a new brushed finish, and ran on-site build trials in China to hold F&P fit-and-finish standards, so the product sits seamlessly alongside the Thai-made F&P range.
- **Completed a dedicated cost-out rotation**, driving cost-down design and warranty-reduction projects across multiple product lines.
- **Owned packaging validation** for international spare-parts shipments, defining cosmetic acceptance criteria and protective specifications for premium components across long-haul freight.

- **Selected for the F&P Future Leaders program;** directed the Product Development Health & Safety team, gaining formal people-leadership experience.

Product Development Engineer

Stallion Plastics, New Zealand | May 2012 - Jun 2017

- **Sole in-house design engineer** producing mechanical designs, manufacturing drawings, and tooling documentation for new products across polymer and metal fabricated components.
- **Collaborated extensively with toolmakers and machinists** to optimise tooling, reduce cost, and improve baseline quality; designed wooden master patterns used to cast aluminium production tooling.
- **Managed part procurement and quality control** across the production pipeline, and supported partner companies with mechanical design services.

SIDE PROJECT

redactcv.com - Founder & Builder (B2B SaaS)

Self-initiated build alongside recruitment work | Mar 2026 - Present

- Identified a high-friction manual workflow (CV anonymisation) and shipped an end-to-end automated product from problem to live beta in two weeks - the same problem-to-shipped-solution loop I bring to hardware.

EDUCATION

Bachelor of Engineering (Honours), Product Development - Massey University, New Zealand